

COMPUTING SPECTRAL PROPERTIES OF DIFFERENTIAL OPERATORS ON SURFACES

GUSTAV CONRADIE*, MATTHEW J. COLBROOK*, AND DANIEL FORTUNATO†

Abstract. Computing spectra of differential operators on surfaces is crucial in many applications, ranging from medical image analysis to nanoscale quantum structures. However, discretization of the infinite-dimensional operator and the underlying surface can lead to inaccurate and misleading results. This paper introduces the **SurfSpec** suite of algorithms to compute eigenvalues and spectral measures of operators on surfaces with error control and convergence guarantees. We combine methods based on complex contour integration and residuals to compute discrete spectra and high-frequency eigenmodes, including of nonlinear eigenvalue problems. We then compute convolutions of spectral measures with rational kernels to deal with continuous spectra. In both cases, the key tool is the operator resolvent applied to functions, which is computed using high-order methods that solve surface PDEs. We illustrate our algorithms on a variety of differential operators and surfaces, including a triceratops.

Key words. spectrum, differential eigenvalue problems, spectral measures, surface PDEs

MSC codes. 35P05, 35P30, 65N25, 65N35

1. Introduction. Given a surface $M \subset \mathbb{R}^3$ and a Hilbert space $\mathcal{H}$ of functions on M , we study the spectral properties of a linear differential operator $\mathcal{L}$ with domain $\mathcal{D}(\mathcal{L}) \subset \mathcal{H}$. The associated eigenvalue problem is

$$(1.1) \quad \mathcal{L}f = \lambda f, \quad f \in \mathcal{D}(\mathcal{L}), \quad \lambda \in \mathbb{C}.$$

For surfaces with boundary, boundary conditions such as Dirichlet or Neumann may be imposed. Since $\mathcal{H}$ is infinite-dimensional, the spectrum $\text{Sp}(\mathcal{L})$, defined formally as

$$(1.2) \quad \text{Sp}(\mathcal{L}) = \{z \in \mathbb{C} : z\mathcal{I} - \mathcal{L} \text{ is not boundedly invertible}\},$$

may include continuous components in addition to eigenvalues, giving rise to a more general and difficult spectral problem than (1.1). Accurately computing the spectra of operators is crucial, as surface spectral problems (SSPs) play a key role in various applications. Eigenvalues and eigenfunctions of the Laplace–Beltrami operator are central to spectral shape analysis, serving as surface fingerprints or “Shape-DNA” [75, 76], and are used for segmentation [74], local registration [43], heat kernel smoothing [78], and constructing geometry-adapted function bases [55]. These methods are widely applied in medical image analysis [64, 77, 82] and computer graphics [12, 96]. Beyond shape analysis, Laplace–Beltrami spectra are used for dimension reduction and data representation [4, 14], and have been applied in modeling cell behavior [67]. Spectral analysis of other differential operators is also important. For instance, the spectrum of a Schrödinger operator with curvature-dependent potential arises in the stability analysis of interfaces in reaction–diffusion systems [1, 44, 45], and in modeling constrained particles [25] or nanoscale quantum structures [29, 31]. Computational tools for computing spectra can also inform conjectures in spectral geometry or, when error control is provided, be used in computer-assisted proofs [51, 65, 66].

SSPs are typically approached via a “discretize-then-solve” framework, where the operator is approximated by a finite matrix and a (potentially generalized) eigenvalue problem is solved. Discretization methods include embedding techniques [10, 54, 59], meshless methods [89, 95], finite elements [57, 76], and finite differences [39, 54]. However,

*Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, CB3 0WA, UK (gjc51@cam.ac.uk, m.colbrook@damtp.cam.ac.uk).

†Center for Computational Mathematics and Center for Computational Biology, Flatiron Institute, New York, NY 10010, USA (dfortunato@flatironinstitute.org).

discretization can introduce issues like spectral pollution (spurious eigenvalues), spectral invisibility (missing true eigenvalues), and failure to capture continuous spectra [8, 9, 23, 56]. These problems often persist even with finer discretizations and are further exacerbated in SSPs due to the added complexity of discretizing the surface.

In this work, we adopt a “solve-then-discretize” framework for SSPs, aiming to delay discretization as much as possible. Rather than discretizing the operator itself, we discretize only the functions it acts on. This is advantageous, as controlling errors in function discretization is generally more tractable than in operator discretization. This methodology has been successfully applied to infinite-dimensional eigenvalue problems (not on surfaces) [23, 49], and also underpins work on semigroups [15], spectral measures [21, 91], spectra [16, 18, 23], continuous Krylov methods [36, 68, 88], and infinite-dimensional QL/QR algorithms [17, 90]. In this paper, we extend three such algorithms to SSPs: **contFEAST** [49], which computes eigenvalues of differential operators, **infBEYN** [23], which solves nonlinear eigenvalue problems, and **SpecSolve** [21], which computes spectral measures of self-adjoint operators.

Our first key tool is the operator resolvent [53, Sec. III.6]:

$$R(z, \mathcal{L}) = (z\mathcal{I} - \mathcal{L})^{-1}, \quad z \notin \text{Sp}(\mathcal{L}).$$

Contour integrals involving the resolvent yield projections onto eigenspaces, enabling accurate eigenvalue computation, and convolution with high-order rational kernels allows computation of spectral measures. We only use applications of the resolvent to functions, so we can avoid the woes of discretization described above and choose the discretization size adaptively depending on z to maintain high accuracy. Additionally, the resolvent norm provides a posteriori error bounds, essential for verifying results.

Our second key tool is the high-order fast direct surface PDE solver in [34], which we use to compute the action of the resolvent by solving shifted linear systems:

$$(1.3) \quad (z\mathcal{I} - \mathcal{L})u = f, \quad u \in \mathcal{D}(\mathcal{L}).$$

This is the only step where discretization occurs. The solver achieves high-order accuracy, is structured to handle multiple right-hand sides efficiently, and supports a broad class of PDEs and surfaces. Moreover, in all of our algorithms the shifted systems for different z can be solved in parallel, enabling effective use of modern computing architectures.

Combining these key tools, we introduce **SurfSpec**, a suite of routines for computing spectral properties of differential operators on surfaces. To the best of our knowledge, this package is the first to enable error-controlled computation of high-frequency eigenmodes for general PDEs on generic smooth surfaces in $\mathbb{R}^3$, as well as spectral measures and solutions to nonlinear eigenvalue problems. All algorithms come with convergence guarantees obtained by combining the results in the papers [21, 23, 49] with the convergence of the high-order solver [34]. Code for **SurfSpec** and all examples in this paper is available at <https://github.com/GustavConradie1/SurfSpec>.

1.1. Organization of the paper. In section 2, we introduce the **surfacefun** PDE solver and outline its key features. Section 3 covers surface differential eigenvalue problems, detailing how resolvent-based contour integral methods can be used to compute eigenvalues with error control, and how this framework extends to nonlinear eigenvalue problems. In section 4, we define spectral measures of linear operators and show how to compute them using resolvent-based convolution methods, including on unbounded surfaces. Each section includes numerical examples demonstrating the accuracy and practicality of the proposed methods.

2. Solving surface PDEs. Let M be a smooth manifold embedded in $\mathbb{R}^3$ that is locally parametrized by a smooth map $\mathbf{x}(\xi, \eta) : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. We begin by briefly discussing how to define derivatives, and hence differential equations, on M . The metric tensor g on M encapsulates its geometry and is given by

$$g = \begin{pmatrix} g_{\xi\xi} & g_{\xi\eta} \\ g_{\eta\xi} & g_{\eta\eta} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_\xi \cdot \mathbf{x}_\xi & \mathbf{x}_\xi \cdot \mathbf{x}_\eta \\ \mathbf{x}_\eta \cdot \mathbf{x}_\xi & \mathbf{x}_\eta \cdot \mathbf{x}_\eta \end{pmatrix},$$

where $\mathbf{x}_\xi$ and $\mathbf{x}_\eta$ represent the partial derivatives of $\mathbf{x}$ with respect to ξ and η respectively. The corresponding inverse metric tensor is

$$g^{-1} = \begin{pmatrix} g^{\xi\xi} & g^{\xi\eta} \\ g^{\eta\xi} & g^{\eta\eta} \end{pmatrix} = \frac{1}{|g|} \begin{pmatrix} g_{\eta\eta} & -g_{\xi\eta} \\ -g_{\eta\xi} & g_{\xi\xi} \end{pmatrix},$$

where $|g| = \det g$. The components of the inverse metric tensor are used to define tangential derivative operators on the surface. In particular, if we define

$$\xi_{\mathbf{x}} = g^{\xi\xi} \mathbf{x}_\xi + g^{\xi\eta} \mathbf{x}_\eta, \quad \eta_{\mathbf{x}} = g^{\xi\eta} \mathbf{x}_\xi + g^{\eta\eta} \mathbf{x}_\eta,$$

then the tangential derivatives in the Cartesian coordinate directions of a scalar field $u = u(\xi, \eta) = u(\mathbf{x}(\xi, \eta))$ on M are given by [35]

$$\partial_x^M u = \xi_x \frac{\partial u}{\partial \xi} + \eta_x \frac{\partial u}{\partial \eta}, \quad \partial_y^M u = \xi_y \frac{\partial u}{\partial \xi} + \eta_y \frac{\partial u}{\partial \eta}, \quad \partial_z^M u = \xi_z \frac{\partial u}{\partial \xi} + \eta_z \frac{\partial u}{\partial \eta}.$$

Familiar quantities can be defined in terms of these tangential derivative operators. For example, the gradient of a scalar field u defined on M is a tangential vector field $\nabla_M u = (\partial_x^M u, \partial_y^M u, \partial_z^M u)$, and the divergence of a vector field $\mathbf{u} = (u_1, u_2, u_3)$ tangent to M is a scalar field $\nabla_M \cdot \mathbf{u} = \partial_x^M u_1 + \partial_y^M u_2 + \partial_z^M u_3$. The surface Laplacian Δ_M , the negative of which is known as the Laplace–Beltrami operator, is

$$\Delta_M u = \nabla_M \cdot \nabla_M u = \partial_x^M \partial_x^M u + \partial_y^M \partial_y^M u + \partial_z^M \partial_z^M u.$$

In this work, we consider general second-order, variable-coefficient elliptic partial differential operators $\mathcal{L}_M$ on M , which take the form

$$\mathcal{L}_M u(\mathbf{x}) = \sum_{i=1}^3 \sum_{j=i}^3 a_{ij}(\mathbf{x}) \partial_i^M \partial_j^M u(\mathbf{x}) + \sum_{i=1}^3 b_i(\mathbf{x}) \partial_i^M u(\mathbf{x}) + c(\mathbf{x}) u(\mathbf{x}),$$

for smooth coefficients a_{ij} , b_i , and c on M , and ∂_1^M , ∂_2^M and ∂_3^M identified with ∂_x^M , ∂_y^M and ∂_z^M respectively; the methods used can be extended straightforwardly to higher-order PDEs. The corresponding surface PDE on M then takes the form

$$(2.1) \quad \mathcal{L}_M u(\mathbf{x}) = f(\mathbf{x}), \quad \mathbf{x} \in M,$$

where $f(\mathbf{x})$ is a smooth function defined on M . If M is a surface with boundary (an open, as opposed to closed, surface), for $u(\mathbf{x})$ to be uniquely determined (2.1) may be accompanied by boundary conditions, e.g., $u(\mathbf{x}) = g(\mathbf{x})$ for $\mathbf{x} \in \partial M$ and some function $g(\mathbf{x})$. For notational convenience in the rest of the paper, we drop the subscripts and superscripts indicating the surface.

To solve problems of the form (2.1), we use a recently developed fast direct solver for surface PDEs based on spectral collocation on unstructured quadrilateral surface meshes. We give a rough overview of the scheme; for more details, see [34]. Let $\{\mathcal{E}_k\}_{k=1}^N$ denote a mesh of N elements each discretized to order p . In particular, each element $\mathcal{E}_k$

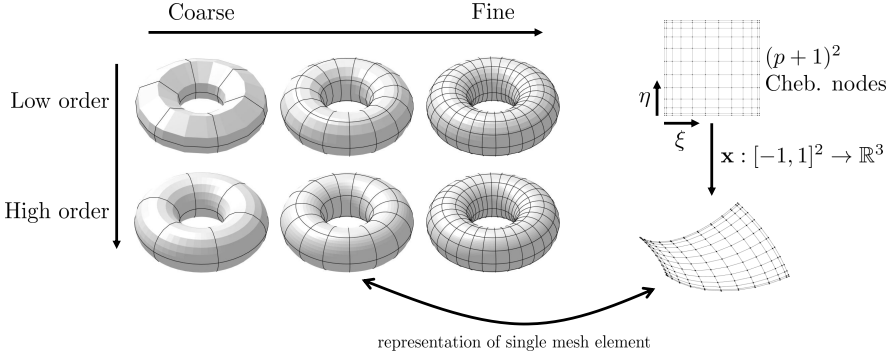


FIGURE 1. *Left: A surface mesh may be refined along two axes. Right: On each element, the surface is sampled at tensor-product Chebyshev nodes of the second kind.*

is represented by sampling M at tensor-product second-kind Chebyshev nodes of order p on $[-1, 1]^2$, resulting in $(p+1)^2$ nodes per element. The function $\mathbf{x} : [-1, 1]^2 \rightarrow \mathbb{R}^3$ which locally parameterizes the surface on each mesh element is then constructed by Chebyshev interpolation. Discretization can be refined either by increasing N (mesh refinement) or p (higher local resolution) (see Figure 1). Local metric tensors are computed via spectral differentiation, and discrete surface differential operators are assembled using matrix algebra to yield a spectral collocation scheme. Functions such as the variable coefficients are sampled at the same Chebyshev nodes on each element.

We split the solution to (2.1) into homogeneous and particular components, and discretize these problems separately on each element, resulting in $2N(p+1)^2 \times (p+1)^2$ linear systems. The solution over the whole surface is then computed in three steps:

1. **Local factorization.** On each mesh element, we perform a local solve to construct solution operators that map Dirichlet boundary data to the homogeneous local solution, along with Dirichlet-to-Neumann maps that yield the corresponding Neumann data. We also compute discrete particular solutions and their associated fluxes.
2. **Global factorization.** In an upward pass, adjacent elements are “glued” to form solution operators, particular solutions, Dirichlet-to-Neumann maps and particular fluxes on the merged parent elements. Each merge enforces continuity of the solution and its binormal derivative along the interface using the Dirichlet-to-Neumann maps and particular fluxes from the child elements.
3. **Solve.** In a downward pass, we begin at the top-level element and use its solution operator, Dirichlet data, and particular fluxes to reconstruct the interface solutions on its child elements. This process continues recursively, reconstructing solutions on all merged interfaces. Once all boundary data is computed for each mesh element, the local solution operators and particular solutions from the factorization step are used to obtain the full solution.

This approach is fast: for fixed p , the local factorization stage takes $\mathcal{O}(N)$ operations, the global factorization stage takes $\mathcal{O}(N^{3/2})$ operations, and the solve stage takes $\mathcal{O}(N \log N)$ operations. Additionally, any boundary data passed to the solver is used only in the solve stage, so we can solve the same PDE with different boundary

Algorithm 3.1 SurfFEAST: Computing eigenvalues of a surface differential operator.

Input: Operator $\mathcal{L}$, quadrature weights $\{w_j\}_{j=1}^\ell$ and nodes $\{z_j\}_{j=1}^\ell$, test functions $F = [f_1 \cdots f_{m+m_0}]$, $\epsilon_{\text{res}}, \epsilon_{\text{tol}} > 0$.

1: **repeat**

2: Solve $(z_k \mathcal{I} - \mathcal{L})G_k = F$ for $k = 1, \dots, \ell$ with relevant boundary conditions.

3: Compute QR decomposition of $(2\pi i)^{-1} \sum_{k=1}^\ell w_k \hat{G}_k$ for orthonormal basis Q .

4: Compute a truncated SVD of $Q \approx \mathcal{U}^* \Sigma \mathcal{V}$, excluding singular values $< \epsilon_{\text{tol}}$.

5: Compute $L = Q^* \mathcal{L} Q$, solve $LX = X\Lambda$ for X and Λ , and update $F = QX$.

6: **until** $\|\mathcal{L}F - F\Lambda\| \leq \epsilon_{\text{res}} \|\Lambda\|$.

Output: Eigenvalues Λ , eigenfunctions $U = QX$.

data without recomputing any factorizations. Updating the right-hand side f is also efficient, requiring only $\mathcal{O}(N \log N)$ steps, as we do not need to recompute solution operators and Dirichlet-to-Neumann maps—just particular solutions and fluxes. The memory complexity also scales at rate $\mathcal{O}(N \log N)$.

We use the open-source MATLAB package `surfacefun` [33], which implements the above solver and provides convenient high-order tools for computing with functions on surfaces. In what follows, it forms a key building block for computing spectral properties of surface operators.

3. Differential eigenvalue problems. This section outlines our approach to computing discrete spectra of differential operators on surfaces. We begin with linear eigenvalue problems and then extend to nonlinear cases, providing error control for both regimes. The core strategy is to evaluate contour integrals involving the resolvent.

3.1. SurfFEAST for linear differential eigenvalue problems. The work [49] provides an analogue of the FEAST algorithm [70] for differential operators on $[-1, 1]$. Motivated by this, we develop an algorithm for solving eigenproblems involving differential operators $\mathcal{L}$ on general surfaces M , which we call SurfFEAST. We assume that the spectrum of $\mathcal{L}$ is discrete in a domain Ω , so any eigenvalue of $\mathcal{L}$ in Ω is isolated from the rest of the spectrum, and that the boundary of Ω does not intersect $\text{Sp}(\mathcal{L})$. The key theoretical tool is the Riesz projector, which is defined for the domain Ω by

$$\mathcal{P}_\Omega = \frac{1}{2\pi i} \int_{\partial\Omega} (z\mathcal{I} - \mathcal{L})^{-1} dz.$$

Under our assumptions, this operator is the projection onto the finite-dimensional invariant subspace spanned by eigenfunctions with eigenvalues in Ω [53, Sec. III.6]. By applying it to suitable test functions, we can recover the corresponding eigenfunctions and eigenvalues inside Ω . Our algorithm for this is summarized in Algorithm 3.1; for the remainder of this section we elucidate the key features of the algorithm.

Assume that the sum of the algebraic multiplicities of eigenvalues inside the contour is m . We select $m + m_0$ test functions $f_1, \dots, f_{m+m_0}$, where m_0 is an oversampling parameter, and let F be the quasimatrix with columns $f_1, \dots, f_{m+m_0}$. We construct F using band-limited random functions¹ as in [32, 49]. Using more test functions than eigenvalues ensures all eigenvalues (counting multiplicity) are captured, improves convergence and helps avoid contamination from nearby eigenvalues just outside the

¹A band-limited periodic random function on a cuboid is given by a truncated Fourier series with coefficients drawn from a Gaussian distribution; the nonperiodic analog (which we use) is constructed by restricting a band-limited periodic random function defined on a large cuboid containing M to the surface M [32, 49].

contour [63, 83]. In practice, m is often unknown but can be estimated via asymptotic formulas like Weyl's law (see subsection 3.3) or stochastic trace estimation [62, 69, 97]; overestimating is preferable. Any excess rank can later be removed via truncated SVD as discussed later in this section; see [85] for details on quasimatrix SVDs.

Given the quasimatrix F , we want to compute $\mathcal{P}_\Omega F$. We approximate the resulting integral via a quadrature rule as follows:

$$\mathcal{P}_\Omega F = \left(\frac{1}{2\pi i} \int_{\partial\Omega} (z\mathcal{I} - \mathcal{L})^{-1} dz \right) F \approx \hat{\mathcal{P}}_\Omega F = \frac{1}{2\pi i} \sum_{k=1}^{\ell} w_k (z_k \mathcal{I} - \mathcal{L})^{-1} F.$$

Here $z_1, \dots, z_\ell$ and $w_1, \dots, w_\ell$ are a set of quadrature nodes and weights, respectively; for the experiments in this paper we use the exponentially convergent trapezoidal rule [87]. For both empirical and theoretical reasons [23], we use medium-sized contours to locate approximate locations of the eigenvalues before using small circular contours to “zoom in” on a particular eigenvalue and compute it to higher accuracy. If $\mathcal{L}$ is self-adjoint, then all of its eigenvalues are real and so we use circular or elliptical contours centered on the real axis, with quadrature nodes taken to lie off the real axis (see (3.4)). If F is real-valued, then

$$(z\mathcal{I} - \mathcal{L})^{-1} F + (\bar{z}\mathcal{I} - \mathcal{L})^{-1} F = 2\text{Re}((z\mathcal{I} - \mathcal{L})^{-1}) F$$

and so we can halve the number of computations required; for example, all Laplace–Beltrami eigenfunctions can be taken to be real-valued and so we may take F real.

Construction of $\hat{\mathcal{P}}_\Omega F$ then requires the solution to ℓ shifted linear surface PDEs, each with $m + m_0$ right-hand sides, i.e.,

$$(3.1) \quad (z_k \mathcal{I} - \mathcal{L}) g_{i,k} = f_i, \quad i = 1, \dots, m + m_0, \quad k = 1, \dots, \ell.$$

The solver discussed in section 2 has the advantage that when solving the PDE for $m + m_0$ different right-hand sides f , we do not need to recompute solution operators and Dirichlet-to-Neumann maps [34]. It also enables a high degree of parallelization: we may solve the shifted linear systems in parallel for each shift, and for each shifted linear system the local factorization step (see section 2) can be carried out in parallel on each mesh element. The solver produces approximate solutions $\hat{g}_{i,k}$ to (3.1). If the quasimatrix with columns $\hat{g}_{1,k}, \dots, \hat{g}_{m+m_0,k}$ is denoted by $\hat{G}_k$, then

$$\hat{\mathcal{P}}_\Omega F = \frac{1}{2\pi i} \sum_{k=1}^{\ell} w_k \hat{G}_k.$$

The columns of $V = \mathcal{P}_\Omega F$ span the subspace of eigenfunctions whose eigenvalues lie inside Ω , and $\hat{V} = \hat{\mathcal{P}}_\Omega F$ approximates this. We compute an orthonormal basis $\{q_j\}_{j=1}^m$ of $\hat{V}$ with respect to the inner product of $\mathcal{H}$ via a QR factorization [85]; we let Q be the quasimatrix with columns q_j . The choice of band-limited periodic functions usually ensures that $\hat{\mathcal{P}}_V F$ has linearly independent and well-conditioned columns when $\mathcal{L}$ is normal [49].

Next, we apply a truncated SVD to Q to compensate for overestimating m . If $m_0 > 0$, the span of the columns of $\hat{V}$ will include approximate eigenfunctions corresponding to eigenvalues which lie outside Ω , as well as (rarely) spurious eigenvalues, and the truncated SVD will remove some of these depending on the tolerance ϵ_{tol} ; typically, singular values drop sharply after those corresponding to true eigenvalues inside the contour. The larger m_0 is, the more eigenvalues outside the contour will be picked up, beginning with the ones closest to the contour. Eigenvalues outside the

contour will typically be approximated inaccurately, with the accuracy decreasing the further outside the contour they are, so it is desirable to remove them.

Finally, we take a Rayleigh-Ritz projection $L = Q^* \mathcal{L} Q \in \mathbb{C}^{n \times n}$. The elements of L are given by the inner products $L_{ij} = \langle q_i, \mathcal{L} q_j \rangle$. This is done in `surfacefun` for the $L^2(M)$ norm using a quadrature rule based on Chebyshev points and weights and using the square root of the Jacobian. Ignoring numerical error, the eigenvalues Λ of L are exactly the eigenvalues of $\mathcal{L}$ that lie inside the given contour [49]; hence, we have reduced the infinite-dimensional eigenvalue problem to a finite-dimensional one without directly discretizing $\mathcal{L}$. The corresponding eigenfunctions are given by the columns of $U = QX$, where X is the matrix of eigenfunctions of L given by $LX = X\Lambda$.

To improve accuracy, we can iterate the algorithm, updating the test functions according to $F = QX$ and repeating the previous steps. We terminate when $\|\mathcal{L}F - F\Lambda\| \leq \epsilon_{\text{res}} \|\Lambda\|$; motivation for this termination condition can be found in the next section. If the contour encircles only a single eigenvalue of multiplicity 1, no benefit is obtained from iterations beyond the first; otherwise, taking two iterations is typically both necessary and sufficient for accurate computations [48]. These iterations can be interpreted in terms of rational subspace iteration [83].

3.2. Error control. Due to the difficulties of computing spectral properties of infinite-dimensional operators numerically, verification is crucial. For a normal operator $\mathcal{L}$, it is known that distance to the spectrum is exactly the inverse of the resolvent norm [26], i.e.,

$$(3.2) \quad \text{dist}(z, \text{Sp}(\mathcal{L})) = \|(z\mathcal{I} - \mathcal{L})^{-1}\|^{-1},$$

where $\|(z\mathcal{I} - \mathcal{L})^{-1}\|^{-1} = 0$ if $z \in \text{Sp}(\mathcal{L})$. Moreover, one can characterize the resolvent norm in terms of the injection modulus [18]

$$\|(z\mathcal{I} - \mathcal{L})^{-1}\|^{-1} = \sigma_{\text{inf}}(z\mathcal{I} - \mathcal{L}), \quad \sigma_{\text{inf}}(\mathcal{L}) = \inf\{\|\mathcal{L}f\| : f \in \mathcal{D}(\mathcal{L}), \|f\| = 1\}.$$

Combining these, we get an a posteriori estimate for the distance to the true spectrum of a proposed eigenvalue λ : if the corresponding normalized proposed eigenvector is f , then the distance of λ to the spectrum is bounded above by

$$(3.3) \quad \text{dist}(\lambda, \text{Sp}(\mathcal{L})) \leq \|(\lambda\mathcal{I} - \mathcal{L})f\|.$$

We refer to $\|(\lambda\mathcal{I} - \mathcal{L})f\|$ as the residual. Using this error bound, we can discard spurious eigenvalues and avoid spectral pollution. This also provides justification for the choice of termination condition in [Algorithm 3.1](#).

Furthermore, if $\mathcal{L}$ is self-adjoint we can use similar arguments to verify the solution to the shifted linear systems in (3.1). In particular, since $\text{Sp}(\mathcal{L}) \subset \mathbb{R}$ it follows by (3.2) that $\|(z\mathcal{I} - \mathcal{L})^{-1}\| = 1/\text{dist}(z, \text{Sp}(\mathcal{L})) \leq 1/|\text{Im}(z)|$ for all z with $\text{Im}(z) \neq 0$, and so

$$(3.4) \quad \|\hat{g}_{i,k} - (z_k\mathcal{I} - \mathcal{L})^{-1}f_i\| \leq \frac{\|(z_k\mathcal{I} - \mathcal{L})\hat{g}_{i,k} - f_i\|}{|\text{Im}(z_k)|}.$$

This allows for adaptive solution of the shifted linear systems to improve accuracy and decrease the eigenvalue residual.

For a nonnormal operator, obtaining bounds for the distance to the spectrum in terms of the resolvent is trickier, though possible [16, Sec. 2.2.2]. Moreover, bounds on the resolvent norm can be interpreted in terms of pseudospectral set inclusions [86], which still provide verification and help avoid spectral pollution. Given $\epsilon > 0$, we define the ϵ -pseudospectrum of $\mathcal{L}$ by

$$\text{Sp}_\epsilon(\mathcal{L}) = \text{Cl}\left(\{z \in \mathbb{C} : \|(z\mathcal{I} - \mathcal{L})^{-1}\|^{-1} < \epsilon\}\right),$$

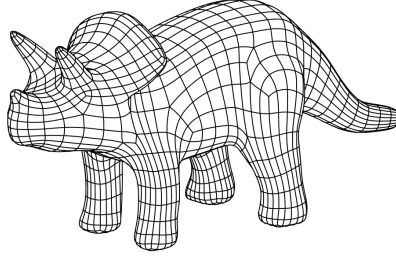


FIGURE 2. The triceratops mesh used in the numerical experiments in [subsection 3.3](#).

where Cl denotes the closure of the set. The pseudospectrum captures how sensitive the spectrum is to perturbations, as shown by the following relation [[86](#), Thm. 4.3]:

$$\text{Sp}_\epsilon(\mathcal{L}) = \text{Cl} \left(\bigcup_{\|E\| < \epsilon} \text{Sp}(\mathcal{L} + E) \right).$$

The pseudospectrum converges to the spectrum as $\epsilon \rightarrow 0$ in a suitable topology which captures spectral pollution and spectral invisibility [[18](#)]. Theoretical bounds on how numerical errors affect the results of contour methods for differential eigenvalue problems, and corresponding convergence results, are typically stated in terms of pseudospectral inclusions [[23](#), [49](#)].

3.3. Example: Laplace–Beltrami and Schrödinger operators. As a first example, we consider the Laplace–Beltrami operator $-\Delta : \mathcal{D}(-\Delta) \rightarrow L^2(M)$ with Dirichlet boundary conditions [[24](#)]. It has a discrete spectrum consisting of eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$, with each eigenvalue having finite multiplicity. For closed surfaces $\lambda_1 = 0$, while for Dirichlet problems on open surfaces $\lambda_1 > 0$ [[13](#)]. We define the eigenvalue counting function $N(\lambda)$ to be the number of eigenvalues less than or equal to λ . Weyl’s law [[92](#)] for surfaces embedded in $\mathbb{R}^3$ states that

$$(3.5) \quad N(\lambda) = \frac{1}{4\pi} \text{area}(M)\lambda + o(\lambda) \quad \text{as } \lambda \rightarrow \infty.$$

We use this as a point of comparison for our numerical computations, and to estimate m . We also compare to the additional conjectured terms in Weyl’s law [[93](#)]:

$$(3.6) \quad N(\lambda) = \frac{1}{4\pi} \text{area}(M)\lambda - \frac{1}{4\pi} \text{length}(\partial M)\lambda^{1/2} + o(\lambda^{1/2}) \quad \text{as } \lambda \rightarrow \infty.$$

Note that for closed surfaces, the second term is 0. A summary of progress towards proving this conjecture can be found in [[50](#)].

As a second example, we compute eigenvalues for surface Schrödinger operators $\mathcal{L} = -\Delta + V(\mathbf{x})$. On compact manifolds, for potentials satisfying a local Lipschitz condition, (3.5) is also satisfied [[72](#)]. We choose a harmonic oscillator potential $V(\mathbf{x}) = \|\mathbf{x}\|^2$. Though it looks simple, for complicated manifolds the potential is far from polynomial in the coordinate charts.

3.3.1. A worked example: a triceratops. We begin with an in-depth example to analyze convergence of [Algorithm 3.1](#) as different parameters are varied. We apply [Algorithm 3.1](#) to the Laplace–Beltrami operator and the Schrödinger operator with harmonic oscillator potential on the triceratops mesh shown in [Figure 2](#). The original mesh was obtained from <https://quaternius.com>, and has been smoothed and simplified

using Rhino 8; the final mesh has 1781 faces and is of order 12. The Python script `surfacepts` converts the SubD mesh exported by Rhino to a `.csv` file that can be imported and used in the `surfacefun` code.

We compute the first 1000 eigenvalues of both the Laplace–Beltrami operator and Schrödinger operator using medium-sized elliptical contours and (over)estimating m based on Weyl’s law. In particular, we used elliptical contours of semi-major axis 10.1 and semi-minor axis 1 which are centered at points $(10(2n + 1), 0)$ for $n \geq 0$, and took $m_0 = 30 > m$. The overlap between different contours is to mitigate any decrease in accuracy near the edge of the contour, as the eigenvalues we are interested in are at least 0.1 away from the edge of the contour and we can compare results from overlapping contours to ensure that no eigenvalues are missed. Further discussion on the choice of contour can be found in [20, 23, 41, 87] and the references contained within. We also use a truncated SVD with $\epsilon_{\text{tol}} = 10^{-8}$ to minimize the effects of overestimating m . Due to the exponential convergence of the quadrature formula, relatively few quadrature nodes are needed. We solve only $l = 15$ shifted linear systems for each contour integral (which, after exploiting symmetry, corresponds to 30 quadrature nodes); subsequent analysis will show that even fewer nodes would suffice.

We also compute the corresponding residuals following (3.3). The residual calculations are helpful to avoid any spurious eigenvalues resulting from not knowing m , inaccurate linear solves, the quadrature rule not having converged or contamination from eigenvalues just outside the contour: 1 eigenvalue across the first 1000 computed for the Laplace–Beltrami operator and 2 eigenvalues for the Schrödinger operator were spurious (having residual much larger than 1). The residuals of the eigenvalues computed for the Laplace–Beltrami operator using the described setup are shown on the top left of Figure 3. We clearly identify the spurious eigenvalue (highlighted in red), and also note that the higher frequency eigenmodes are harder to accurately resolve and hence have larger residuals. If we use circular contours with the same centres and radius 10.1 instead, the eigenvalues and residuals obtained are almost the same (the computed residuals typically agree to 8 decimal places) but there are 3 and 10 spurious eigenvalues for the Laplace–Beltrami and Schrödinger operators respectively.

Unlike for many “discrete-then-solve” schemes, where spurious eigenvalues can persist even as large discretization sizes are taken [23, 56], for our contour methods the spurious eigenvalues can be easily identified by residuals and disappear for even slight improvements of the discretization size. For example, for the spurious eigenvalue of the Laplace–Beltrami operator $\lambda_{\text{spurious}} \approx 578.91$, if we increase l from 15 to 20, reduce m by the number of computed eigenvalues outside the contour (but still including the spurious eigenvalue) or equivalently decrease ϵ_{tol} from 10^{-8} to 10^{-6} , or adjust the centre of the contour to be closer to the spurious eigenvalue, then the algorithm no longer outputs the spurious eigenvalue. This is not the case for increases of p , indicating that the spurious eigenvalue arises due to inaccurate quadrature rules or overestimating m rather than inaccuracy in the solves. The same observations hold for the two spurious eigenvalues outputted by the algorithm applied to the Schrödinger operator. By using residuals to identify spurious eigenvalues, we can adaptively tune parameters to obtain accurate results.

We can also “zoom in” on the individual eigenvalues using a circular contour to compute them to higher accuracy. Restricting to the 100th and 1000th eigenvalue, we vary the number of quadrature nodes and the refinement of the mesh to see how convergence is affected. Throughout, residuals are smaller for the lower frequency eigenvalue. The bottom panels of Figure 3 show convergence as we vary the number

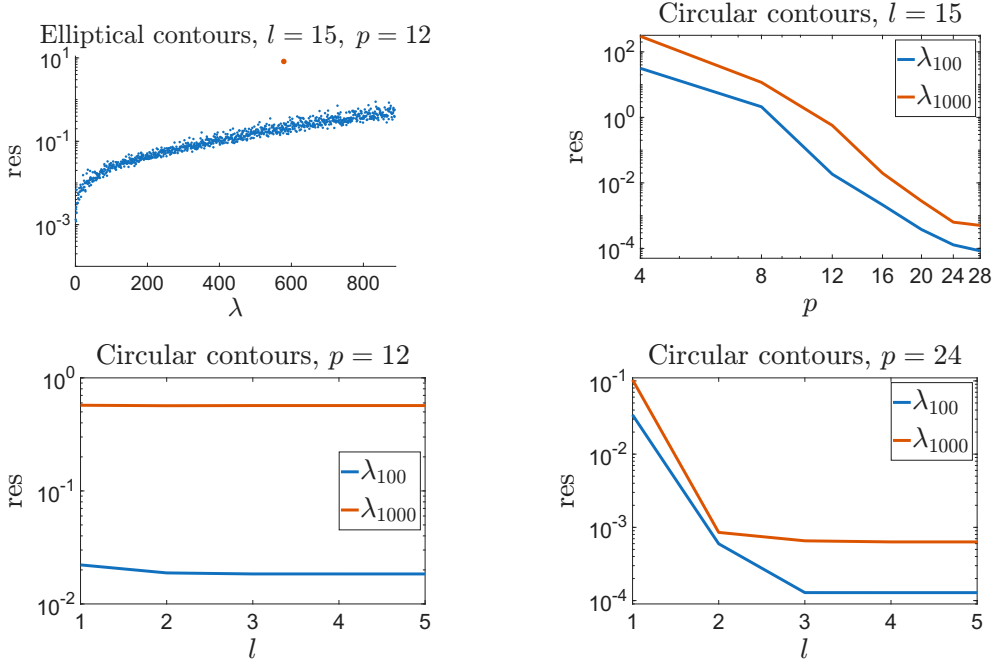


FIGURE 3. Top left: The residuals of the first 1000 eigenvalues of the Laplace-Beltrami operator on a triceratops mesh computed using elliptical contours; the only spurious eigenvalue is highlighted in red. Top right: The residuals of the 100th and 1000th Laplace-Beltrami eigenvalues on a triceratops mesh with varying order using circular contours with $l = 15$. Bottom left: The residuals of the 100th and 1000th eigenvalues on a coarse triceratops mesh using circular contours with different numbers of quadrature nodes. Bottom right: The residuals of the 100th and 1000th eigenvalues on a fine triceratops mesh using circular contours with different numbers of quadrature nodes.

of quadrature nodes for surfaces of order 12 and 24 respectively; we see that the convergence is rapid, with residuals quickly stagnating even after only using 2×3 quadrature nodes. Slightly more quadrature nodes are needed for convergence for the higher-order surface; this is an error arising from coarseness of the surface no longer dominates the quadrature error. As expected, refining the mesh by increasing the order p does improve accuracy and decrease the residual; recall that the order is adjusted by varying the number of Chebyshev nodes on each patch. This is shown clearly on the right of Figure 3; we use $l = 15$ nodes throughout to ensure that quadrature errors are not affecting the eigenvalues or residuals. The rate of decrease appears to stagnate as we increase p ; this is a restriction of p -refinement for the solver used, as the discretized geometries considered are only approximately smooth [34]. In Figure 7 we consider h -refinement and a linear (on a log-log plot) relation between the residual and number of mesh elements is observed.

3.3.2. Further examples. To demonstrate the versatility of our code, we apply it to six different surfaces with both known and unknown spectra:

- *Sphere*: the unit 2-sphere has eigenvalues $l(l+1)$ for l a non-negative integer with multiplicity $2l+1$, with eigenfunctions the spherical harmonics [5];
- *Hemisphere*: the 2-hemisphere with Dirichlet boundary conditions has eigenvalues given by $l(l+1)$ for l a positive integer, with multiplicity l [3];
- *Blob*: a unit sphere deformed by a smooth, random function;
- *Torus*: the surface of revolution of $(x-2)^2 + z^2 = 1$ around the y -axis;

- *Möbius strip*: the parametrization obtained by sweeping out a line segment; this surface is non-orientable;
- *Triceratops*: the imported mesh as discussed in the previous subsection.

For surfaces with boundary, in each case we use Dirichlet boundary conditions. As before, the Laplace–Beltrami operator and Schrödinger operator with harmonic oscillator potential on each of these surfaces has nonnegative, real eigenvalues, so we can systematically compute the eigenvalues by taking elliptical contours centered on the real axis; by doing so, we compute the first 1000 eigenvalues, eigenfunctions and residuals of the Laplace–Beltrami and Schrödinger operators. Since both operators are self-adjoint, the residuals bound the distance to the true eigenvalue of the infinite-dimensional operator (see [subsection 3.2](#)). The parameters are chosen adaptively to ensure small residuals and m is (over)estimated based on Weyl’s law; for example, for the triceratops we used the parameters described in [subsection 3.3.1](#). We then “zoom in” on the 1000th such eigenvalues for each surface/operator pair, taking $m_0 = m = 1$ and refining the surface to produce small residuals. The eigenvalues, eigenfunctions and residuals are shown in [Figures 4 and 5](#); for eigenfunctions with multiplicity, we display the eigenfunction with the largest residual to show the worst-case problem. Note that for the sphere and hemisphere we require a larger number of quadrature nodes than usual due to the high multiplicity. We do not show the results for the Schrödinger operator on the sphere and hemisphere, as harmonic oscillator potential is constant on these surfaces.

We also compare the eigenvalue counting function $N(\lambda)$ as computed by our algorithm to the asymptotic expansion $N^w(\lambda)$ given by Weyl’s law [\(3.5\)](#); for open surfaces, we compare to both [\(3.5\)](#) and [\(3.6\)](#). We see that there is strong agreement and as λ increases, the relative error between the two quantities decreases, as the asymptotic expansion becomes more accurate for large λ . The error decreases faster for open surfaces when we include the second term in Weyl’s law for the Laplace–Beltrami operator, as expected, and also for the Schrödinger operator on the Möbius strip. For the sphere and hemisphere, the plotted Weyl approximation and the discrete count coincide, leading to the observed machine-precision dips.

3.4. Nonlinear differential eigenvalue problems. Contour-integral-based methods are also effective for nonlinear eigenvalue problems (NLEPs). For example, recent work [\[23\]](#) has generalized Beyn’s method [\[6\]](#). We consider the eigenvalue problem

$$(3.7) \quad T(\lambda)f = 0, \quad T(\lambda) : \mathcal{D}(T) \rightarrow \mathcal{H}, \quad f \in \mathcal{D}(T),$$

where T may depend nonlinearly on λ and $\mathcal{D}(T) \subset \mathcal{H}$ is dense. The spectrum of T , $\text{Sp}(T)$, is the set of points $\lambda \in \mathbb{C}$ such that $T(\lambda) : \mathcal{D}(T) \rightarrow \mathcal{H}$ is not boundedly invertible. Fix an open subset $\Omega \subset \mathbb{C}$. We make the following technical assumptions:

- for each $\lambda \in \Omega$, $T(\lambda)$ is a Fredholm operator;
- for each $u \in \mathcal{D}(T)$, $\lambda \mapsto T(\lambda)u$ is holomorphic;
- there exists at least one point in Ω not in the spectrum $\text{Sp}(T)$.

Under these conditions the spectrum is discrete in Ω [\[61, Thm. 1.3.1\]](#). For example, the nonlinear eigenvalue problem [\(3.9\)](#) with Dirichlet boundary conditions on the Möbius strip satisfies these conditions: $T(\lambda)$ is Fredholm for every λ by elliptic PDE theory [\[84, Sec. 5.1\]](#), $\lambda \mapsto T(\lambda)u$ is holomorphic in λ for any $u \in \mathcal{D}(T)$, and by considering a relevant variational principle it can be shown that the NLEP only has real eigenvalues [\[40, Sec. 3\]](#) and so any elliptical or circular domain Ω satisfies the third condition.

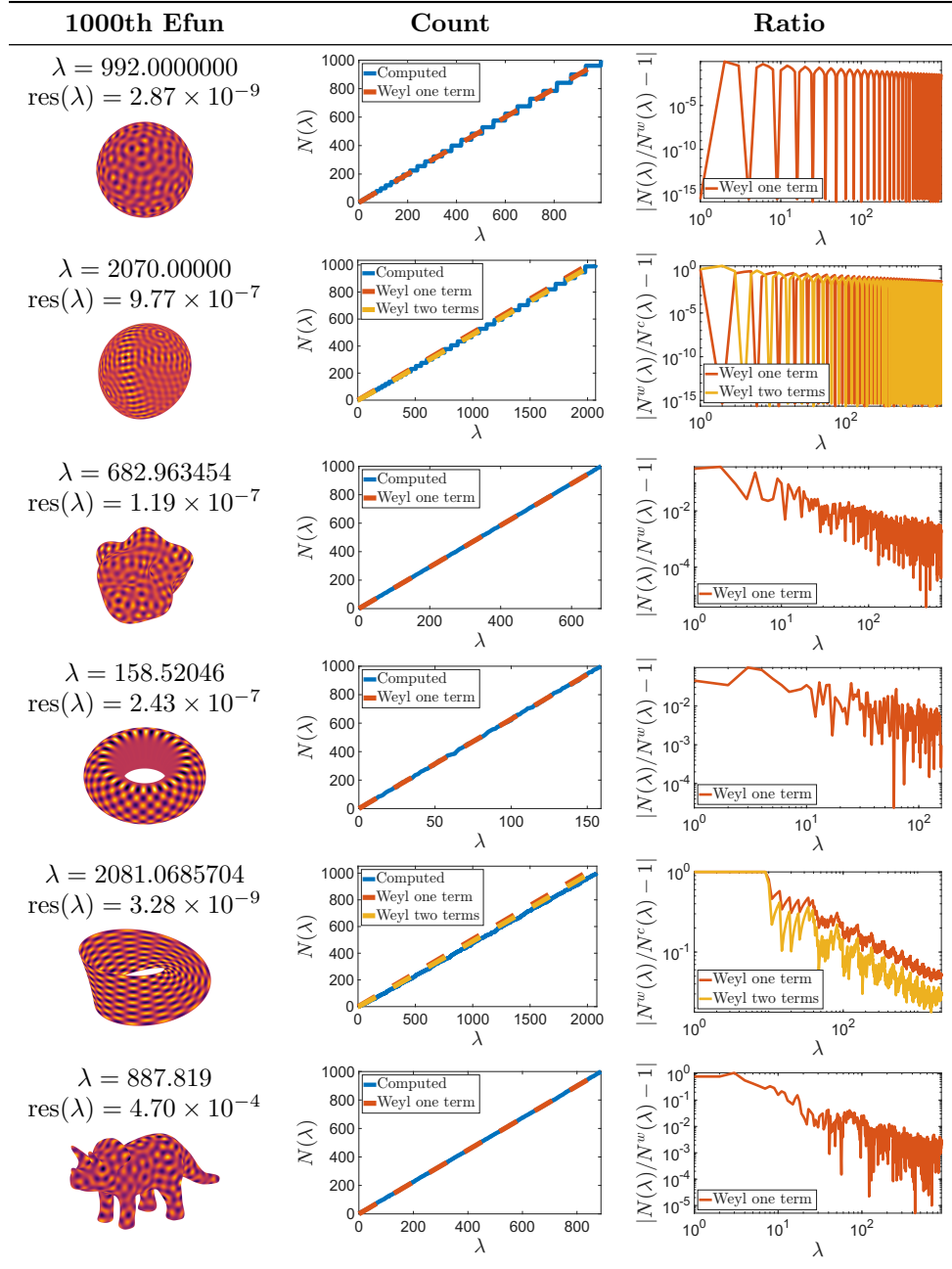


FIGURE 4. Comparison of computed Laplace–Beltrami eigenvalues with Weyl asymptotics, together with the corresponding high-frequency eigenfunctions.

The key theoretical tool in applying contour integration methods is Keldysh’s theorem [40]. If m is the sum of the algebraic multiplicities of the eigenvalues inside Ω , then

$$T(z)^{-1} = A(zI - J)^{-1}B^* + R(z) \quad \forall z \in \Omega \setminus \text{Sp}(T),$$

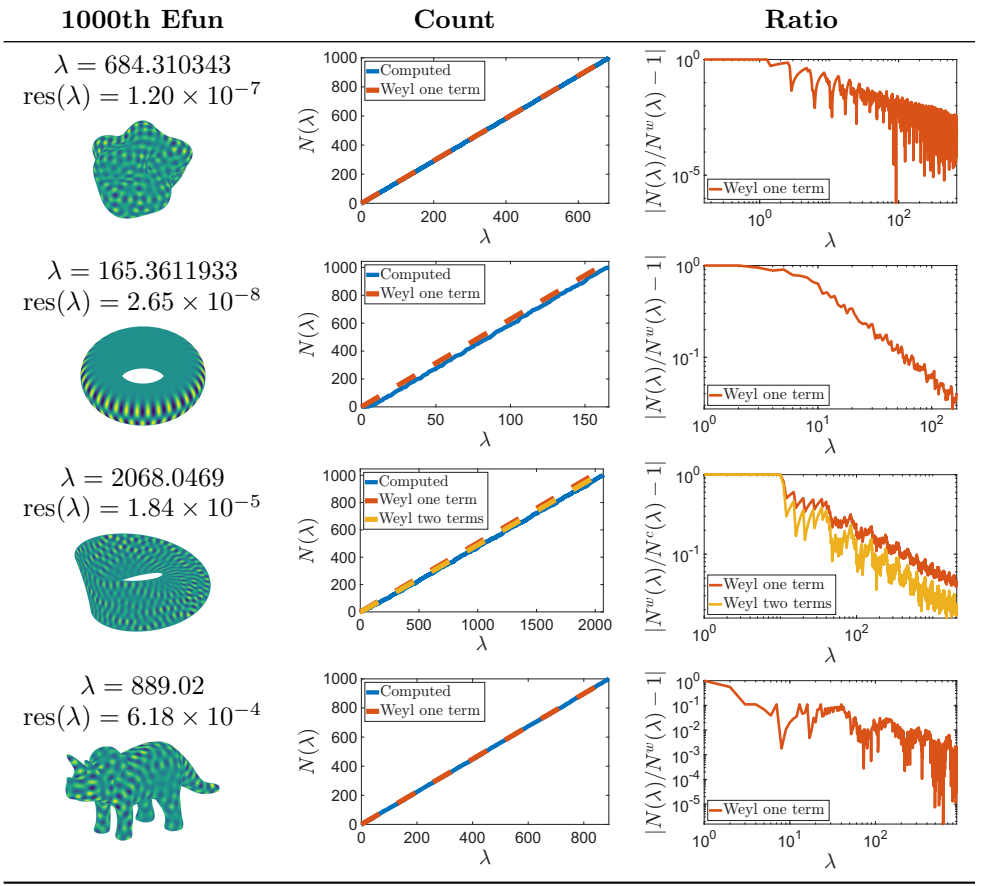


FIGURE 5. Comparison of computed Schrödinger eigenvalues with Weyl asymptotics, together with the corresponding high-frequency eigenfunctions.

where $J \in \mathbb{C}^{m \times m}$ is the Jordan canonical form for the eigenvalues inside Ω , A and B are the normalized canonical Jordan chains and $R(z)$ is a holomorphic remainder.

By Cauchy's theorem, for any holomorphic function f we have that

$$(3.8) \quad \frac{1}{2\pi i} \int_{\partial\Omega} f(z) T(z)^{-1} dz = Af(J)B^*.$$

Using these theoretical results, we develop a solver for nonlinear surface eigenvalue problems, **SurfBeyn**, based on [23]. As for **SurfFEAST**, the key idea is to relate the eigenvalues of (3.7) inside Ω to a finite-dimensional eigenvalue problem using contour integrals. We define

$$A_0 = \frac{1}{2\pi i} \int_{\partial\Omega} T(z)^{-1} dz, \quad \text{and} \quad A_1 = \frac{1}{2\pi i} \int_{\partial\Omega} z T(z)^{-1} dz.$$

As in subsection 3.1, we approximate A_0 and A_1 by applying them to test functions and using a quadrature rule. If we then take a (truncated, if we do not know m exactly) SVD of our approximation $\hat{A}_0 \approx \mathcal{U}\Sigma\mathcal{V}^*$, then the generalized eigenvalue problem $\mathcal{U}^* \hat{A}_1 \mathcal{V} X = \Sigma X \Lambda$ gives the eigenvalues of $T(\lambda)$ inside Ω up to error arising from the

Algorithm 3.2 SurfBeyn: Computing eigenvalues of a nonlinear surface differential operator.

Input: Nonlinear operator T , quadrature weights $\{w_j\}_{j=1}^\ell$ and nodes $\{z_j\}_{j=1}^\ell$, test functions $F = [f_1 \cdots f_{m+m_0}]$, $\epsilon_{\text{res}}, \epsilon_{\text{tol}} > 0$.

- 1: **repeat**
- 2: Solve $T(z_k)G_k = F$ for $k = 1, \dots, \ell$ with relevant boundary conditions.
- 3: Compute $\hat{A}_0 = (2\pi i)^{-1} \sum_{k=1}^\ell w_k \hat{G}_k$ and $\hat{A}_1 = (2\pi i)^{-1} \sum_{k=1}^\ell w_k z_k \hat{G}_k$.
- 4: Compute a truncated SVD of $\hat{A}_0 \approx \mathcal{U}\Sigma\mathcal{V}^*$, excluding singular values $< \epsilon_{\text{tol}}$.
- 5: Form $B = \mathcal{U}^* \hat{A}_1 \mathcal{V} \Sigma^{-1}$ and solve $BX = X\Lambda$ for X and Λ .
- 6: Set $U = \mathcal{U}X$, normalize the columns of U , and update $F = U$.
- 7: **until** $\|T(\lambda_j)u_j\| \leq \epsilon_{\text{res}} \max\{1, |\lambda_j|\}$ for each computed eigenpair (λ_j, u_j) .

Output: Eigenvalues Λ , eigenfunctions $\mathcal{U}X$.

linear solves, quadrature rule and oversampling m .² The algorithm is summarized in [Algorithm 3.2](#). While the residuals $\|T(\lambda)f\|$ do not bound the distance to the spectrum as in (3.3), we can use pseudospectral inclusions to verify computations; the pseudospectrum for nonlinear eigenvalue problems is defined as in [subsection 3.2](#) but with $(z\mathcal{I} - \mathcal{L})^{-1}$ replaced by $T(z)^{-1}$, and analogous results for its construction in terms of residuals and convergence to the spectrum as $\epsilon \downarrow 0$ hold [23].

3.4.1. Example. We consider an example inspired by [47], arising from the partial delay differential equation (PDDE) with Dirichlet boundary conditions:

$$\begin{aligned} u_t(\mathbf{x}, t) &= \Delta u(\mathbf{x}, t) + a(\mathbf{x})u(\mathbf{x}, t) + b(\mathbf{x})u(\mathbf{x}, t-1), & \mathbf{x} \in M, t \geq 0, \\ u(\mathbf{x}, t) &= 0, & \mathbf{x} \in \partial M, t \geq 0. \end{aligned}$$

Taking a Laplace transform with respect to the time variable, we obtain a NLEP of the form (3.7) with

$$(3.9) \quad T(\lambda) = \Delta + (a(\mathbf{x}) - \lambda + e^{-\lambda}b(\mathbf{x}))\mathcal{I}.$$

We consider this problem on the Möbius strip, with $a(\mathbf{x}) = \sin^2(x)\sin^2(y)\sin^2(z)$ and $b(\mathbf{x}) = \sin(x+y+z) + 1.5$, and apply Dirichlet boundary conditions. The Möbius strip mesh used has order $p = 16$ and 210 mesh elements.

As before, we first use elliptical contours to localize the eigenvalues, followed by small, isolated, circular contours to compute them accurately. The eigenvalues are shown in [Table 1](#) along with the corresponding residuals, which allow for interpretations in terms of pseudospectral inclusions and convergence to the true spectrum as in [subsection 3.2](#). Residuals remain small, with the expected drop in accuracy for larger eigenvalues as observed in [subsection 3.3.1](#). [Figure 5](#) displays the 1st, 10th, and 100th eigenfunctions, showing the expected increase in oscillations with eigenvalue magnitude.

We also analyze convergence of the algorithm. First we vary the number of quadrature nodes while keeping the mesh fixed; on the left of [Figure 7](#) we see that while more nodes are needed for convergence than for SurfFEAST as in [Figure 3](#), still only around 25 nodes are required (here we have no symmetry consideration to exploit). To explore how refining the mesh affects convergence, we vary the number of mesh

²This follows by Keldysh's theorem. Assume that we have A_0 and A_1 exactly, and that we know m . Then by (3.8), $A_0 = AB^*$ and $A_1 = AJB^*$ for quasimatrices A and B . Hence the eigenvalue problem $\mathcal{U}^*A_1\mathcal{V}X = \Sigma X\Lambda$ can be written as $(\mathcal{U}^*AJB^*\mathcal{V})X = (\mathcal{U}^*AB^*\mathcal{V})X\Lambda$ and so has the same eigenvalues as J .

Number	Eigenvalue	Residual
1	-1.359074970	6.10×10^{-11}
2	-1.59561416	1.03×10^{-9}
3	-1.85101635	6.69×10^{-11}
4	-2.089752672	5.95×10^{-11}
5	-2.291412643	5.94×10^{-11}
10	-2.835581355	9.27×10^{-11}
20	-3.55705229	8.89×10^{-11}
50	-4.31588030	3.96×10^{-10}
100	-5.0004860	1.40×10^{-8}

TABLE 1

Eigenvalues and corresponding residuals (for pseudospectral inclusions) for the NLEP (3.9).

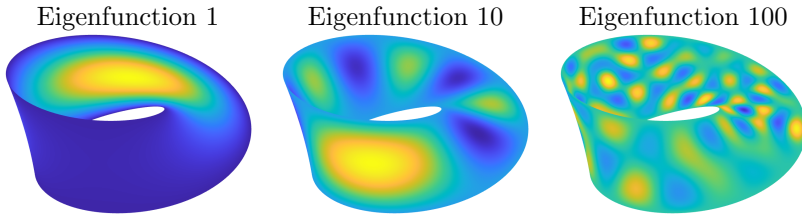


FIGURE 6. *Eigenfunctions of the NLEP (3.9) on the Möbius strip mesh with $p = 16$ and 210 mesh elements using $l = 50$ quadrature nodes.*

elements rather than the order of the mesh as in subsection 3.3.1 (recall Figure 1), using 50 nodes throughout. We see on the right of Figure 7 that the residual varies linearly (on the log-log plot) with the number of elements, suggesting polynomial convergence.

4. Spectral measures on surfaces. Many infinite-dimensional linear operators in physics and related fields have spectra that are not purely discrete, so their eigenfunctions do not form a basis of the Hilbert space. In particular, an operator $\mathcal{L}$ on $\mathcal{H}$ may have non-empty *continuous spectrum*. However, the spectral theorem for self-adjoint operators provides a resolution: projections onto eigenspaces are replaced by a projection-valued measure $\mathcal{E}$ [73, Thm. VIII.6], which assigns an orthogonal projection to each Borel-measurable set such that

$$f = \left(\int_{\mathbb{R}} d\mathcal{E}(\lambda) \right) f, \quad \forall f \in \mathcal{H}, \quad \text{and} \quad \mathcal{L}f = \left(\int_{\mathbb{R}} \lambda d\mathcal{E}(\lambda) \right) f, \quad \forall f \in \mathcal{D}(\mathcal{L}).$$

Thus $\mathcal{E}$ decomposes $\mathcal{H}$ and diagonalizes $\mathcal{L}$. The spectral measure of $\mathcal{L}$ with respect to $f \in \mathcal{H}$ is a scalar measure defined by $\mu_f(S) = \langle \mathcal{E}(S)f, f \rangle$, where $S \subset \mathbb{R}$ is a Borel-measurable set. Such measures are central in applications ranging from quantum mechanics [42, 73] and plasma physics [94] to stochastic processes [38, 52] and signal processing [80]. A similar spectral theorem holds for normal operators.

For a self-adjoint matrix, the spectral measure will consist of Dirac delta functions located at the eigenvalues of the matrix along the real line, with weights depending on the function f . However, infinite-dimensional operators can have continuous spectrum, as in the left of Figure 8 and the right of Figure 10. When $\mathcal{L}$ has a non-empty continuous spectrum, we construct smoothed approximations of μ_f . In particular, we compute a smooth function μ_f^ϵ , with parameter $\epsilon > 0$, that converges weakly to μ_f ,

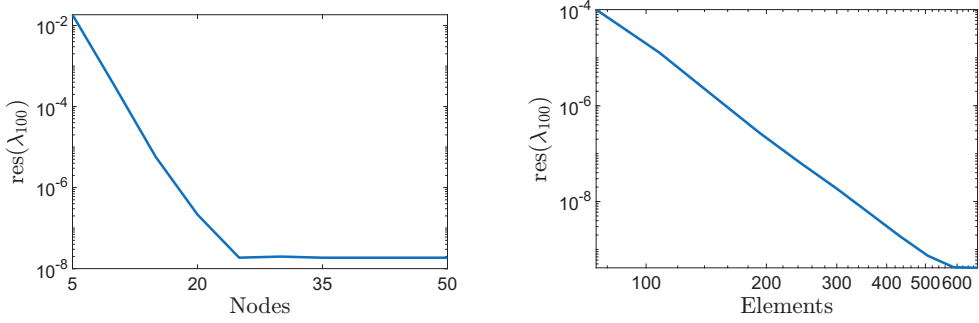


FIGURE 7. *Left: The residual of the 100th eigenvalue of the NLEP (3.9) on a Möbius strip solved using different numbers of quadrature nodes. Right: The residual of the 100th eigenvalue of the same problem solved using varying numbers of mesh elements.*

i.e., for any bounded, continuous function ϕ [7, Ch. 1],

$$\int_{\mathbb{R}} \phi(\lambda) \mu_f^\epsilon(\lambda) d\lambda \rightarrow \int_{\mathbb{R}} \phi(\lambda) d\mu_f(\lambda), \quad \text{as } \epsilon \downarrow 0.$$

To achieve this, we again use the resolvent $(\mathcal{L} - z\mathcal{I})^{-1}$. In particular, an argument using Stone's formula [81] gives a smooth approximation of μ_f in terms of a convolution with the Poisson kernel:

$$(4.1) \quad \mu_f^\epsilon(\lambda) := -\frac{1}{\pi} \text{Im} \left(\langle (\mathcal{L} - (\lambda - i\epsilon)\mathcal{I})^{-1} f, f \rangle \right) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{\epsilon d\mu_f(y)}{\epsilon^2 + (\lambda - y)^2}$$

The function μ_f^ϵ converges weakly to the spectral measure μ_f as $\epsilon \downarrow 0$. Moreover, if μ_f has sufficient local regularity around the point λ , then $\mu_f^\epsilon(\lambda)$ converges to the Radon–Nikodym derivative $\rho_f(\lambda)$ as $\epsilon \downarrow 0$. However, for small ϵ , solving the shifted linear systems to evaluate the resolvent becomes computationally expensive, as we need a large discretization size (see (3.4)). Moreover, the approximation error of (4.1) in ϵ is $\mathcal{O}(\epsilon \log(1/\epsilon))$. These two properties limit the practicality of the Poisson kernel.

4.1. High-order smoothing kernels. Instead of the Poisson kernel, using higher order kernels in the above analysis improves convergence; in particular, since we can use larger ϵ for a specified accuracy, we may use smaller discretization sizes and hence have faster solves. We define

$$K(x) = \frac{1}{2\pi i} \sum_{j=1}^m \left[\frac{\alpha_j}{x - a_j} - \frac{\overline{\alpha_j}}{x - \overline{a_j}} \right], \quad \text{and} \quad K_\epsilon(x) = K(x/\epsilon)/\epsilon,$$

where the $\{a_j\}_{j=1}^m$ are distinct points in the upper half plane. Following [21], we use equispaced poles $a_j = 2j/(m+1) - 1 + i$. We choose the α_j to satisfy the Vandermonde system

$$(4.2) \quad \begin{pmatrix} 1 & \dots & 1 \\ a_1 & \dots & a_m \\ \vdots & \ddots & \vdots \\ a_1^{m-1} & \dots & a_m^{m-1} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_m \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

This choice ensures that the kernel is an m th order kernel. Namely, K integrates to one, its first to $(m-1)$ th moments vanish and $K(x) = \mathcal{O}(1/(1+|x|)^{m+1})$ as $|x| \rightarrow \infty$.

Algorithm 4.1 SurfMeas: Computing spectral measures of self-adjoint differential operators on surfaces.

- 1: **Input:** $\mathcal{L} : \mathcal{D}(\mathcal{L}) \rightarrow \mathcal{H}$ self-adjoint, $f \in \mathcal{H}$, $\lambda \in \mathbb{R}$, $a_1, \dots, a_m \in \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ and $\epsilon > 0$.
 - 2: Solve the Vandermonde system (4.2) for the residues α_j , $1 \leq j \leq m$.
 - 3: Solve $(\mathcal{L} - (\lambda - \epsilon a_j)\mathcal{I})g_j = f$ for $1 \leq j \leq m$ for solutions $\hat{g}_j$.
 - 4: Compute $\mu_f^\epsilon(\lambda) = -\text{Im} \left(\sum_{j=1}^m \alpha_j \langle \hat{g}_j, f \rangle \right) / \pi$.
 - 5: **Output:** $\mu_f^\epsilon(\lambda)$.
-

Since K is a *rational* kernel, we can obtain a new approximation for the spectral measure directly through the resolvent as

$$\mu_f^\epsilon(\lambda) = [K_\epsilon * \mu_f](\lambda) = -\frac{1}{\pi} \text{Im} \left(\sum_{j=1}^m \alpha_j \langle (\mathcal{L} - (\lambda - \epsilon a_j)\mathcal{I})^{-1} f, f \rangle \right).$$

This approximation generalizes (4.1) and can be thought of as an m th order version of Stone's formula. Again, this approximation converges weakly to μ_f . However, now with sufficient regularity conditions on μ_f it converges to ρ_f , both pointwise and in the L^p sense, with a rate of convergence $O(\epsilon^m \log(1/\epsilon))$: a significant increase [21].

In terms of practical implementation, the only requirements are solving the shifted linear systems $(\mathcal{L} - (\lambda - \epsilon a_j)\mathcal{I})g_j = f$ for $j = 1, \dots, m$ and taking inner products of functions on surfaces, both of which have been discussed previously and can be done within the framework of `surfacefun`. This leads to Algorithm 4.1 for computing spectral measures of operators on surfaces, which we call **SurfMeas**.

4.2. Examples. We consider two examples. The first is on a bounded surface, whereas the second is on an unbounded surface.

4.2.1. Internal waves. A common source of continuous spectra of linear operators on compact surfaces is linear pencil problems. We compute the spectra of the linear pencil $\mathcal{A} - \lambda\mathcal{B}$ for $\mathcal{A}$ and $\mathcal{B}$ self-adjoint and $\mathcal{B}$ positive and invertible. We study this through the operator $\mathcal{B}^{-1}\mathcal{A}$. We define the Hilbert space $\mathcal{H}_\mathcal{B}$ by equipping $\mathcal{D}(\mathcal{B}^{1/2})$ with the norm $\|f\|_\mathcal{B} = \sqrt{\langle \mathcal{B}^{1/2}f, \mathcal{B}^{1/2}f \rangle}$ and define a symmetric closed operator $\mathcal{L} = \mathcal{B}^{-1}\mathcal{A}|_{\mathcal{D}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}^{1/2})}$. Under suitable assumptions on $\mathcal{A}$ and $\mathcal{B}$, $\mathcal{L}$ is self-adjoint [19], and so we apply the approach of Algorithm 4.1 to compute its spectral measures. For simplicity, we assume that $f \in \mathcal{D}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}^{1/2})$ which can be a strict (but dense) subspace of $\mathcal{H}_\mathcal{B}$. Our expression for the convolution now becomes

$$\mu_f^\epsilon(\lambda) = [K_\epsilon * \mu](x) = -\frac{1}{\pi} \text{Im} \left(\sum_{j=1}^m \alpha_j \langle (\mathcal{A} - (\lambda - \epsilon a_j)\mathcal{B})^{-1} \mathcal{B}f, \mathcal{B}f \rangle \right),$$

so we must solve $(\mathcal{A} - (\lambda - \epsilon a_j)\mathcal{B})g_j = \mathcal{B}f$ for $j = 1, \dots, m$.

We consider an example inspired by internal waves [30] and take

$$\mathcal{A} = -\partial_{yy} + \cos^2(10x), \quad \mathcal{B} = -\Delta + \mathcal{I}$$

on the torus. Here $\mathcal{H}_\mathcal{B} = H^1(M)$, the Sobolev space of square-integrable functions with square-integrable first (weak) derivatives. The spectra of internal waves are a significant topic in oceanography and the study of rotating fluids [58, 79]. In this example, the spectrum of $\mathcal{L}$ is $[0, 1]$, but the spectral type is unknown [71]. As our

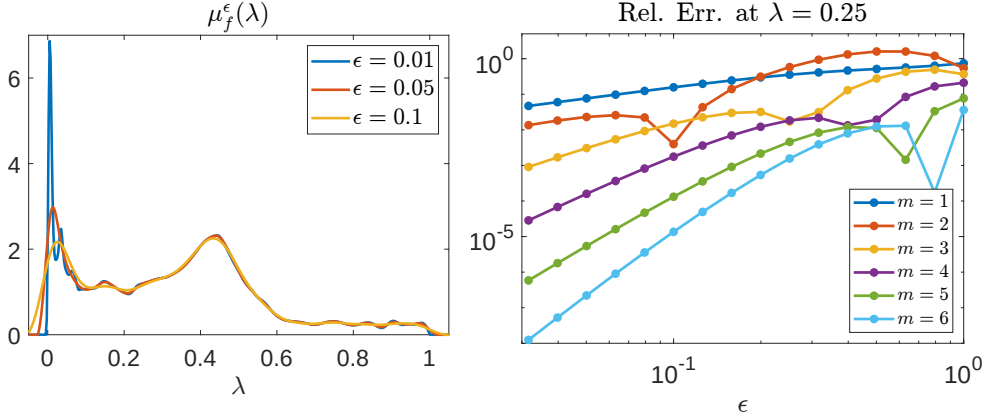


FIGURE 8. Left: Smoothed spectral measures for different choices of ϵ computed using the sixth-order kernel. As $\epsilon \downarrow 0$ these approximations converge weakly to the spectral measures. Right: Convergence to the Radon–Nikodym derivative at $\lambda = 0.25$. The errors are computed by comparing to a value computed at $\epsilon = 0.01$. Note the vastly improved convergence rates for the high-order kernels, with the m th order kernel having m th order convergence, up to logarithmic factors.

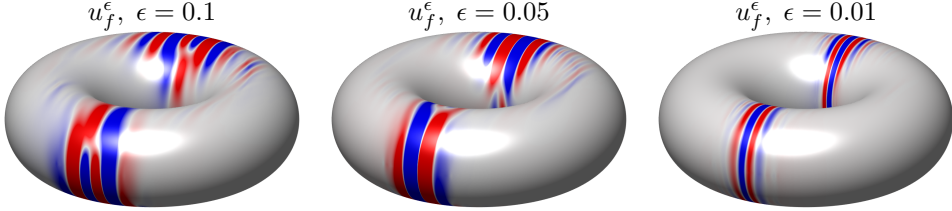


FIGURE 9. The functions in (4.3) for $\lambda = 1$ and various ϵ . As $\epsilon \downarrow 0$, we observe increased oscillations and localization corresponding to a singular generalized eigenfunction. The mesh has order $p = 24$ and 768 mesh elements.

test function, we take

$$f(x, y, z) = \sin(2(x - y + z)^2).$$

Figure 8 (left) shows the smoothed spectral measures computed using the sixth order kernel for various ϵ . As $\epsilon \downarrow 0$, the approximations appear to converge to a density function, suggesting the spectrum of this operator is absolutely continuous. Figure 8 (right) shows the convergence to the Radon–Nikodym derivative at the point $\lambda = 0.25$ and the improved convergence rates of the higher order kernels.

We can also consider the functions

$$(4.3) \quad u_f^\epsilon(x, y, z) = -\frac{1}{\pi} \operatorname{Im} \left(\sum_{j=1}^m \alpha_j g_j(x, y, z) \right),$$

which correspond to the smoothed spectral projection density applied to f . Under suitable conditions, these functions converge in the sense of distributions to generalized eigenfunctions of the operator [22]. Figure 8 shows these functions for $\lambda = 1$ and various ϵ , using a torus mesh with 768 elements and order 24. As $\epsilon \downarrow 0$, we observe increasing oscillations and localization of these functions around a narrow band of the torus. In particular, while approximations converge in a weak sense, they do not converge in $\mathcal{H}_B$.

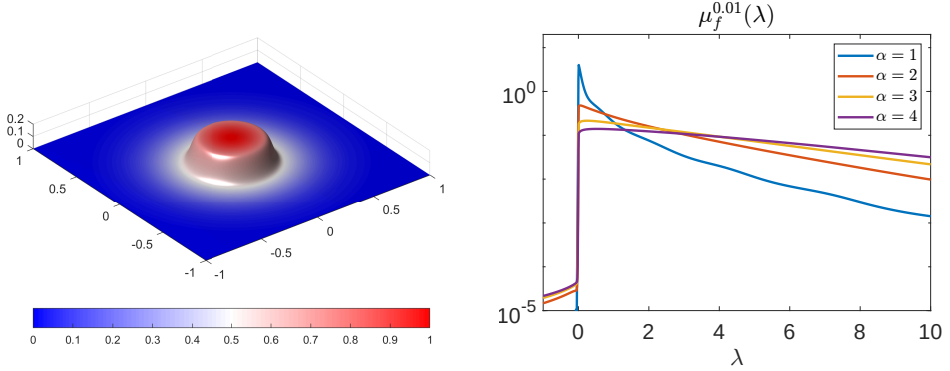


FIGURE 10. *Left: Potential $V(x, y, z) = \exp(-5(x^2 + y^2))$ on the infinite surface. Right: Spectral measures for different α . Larger α leads to more decay in physical space but less in spectral space.*

4.2.2. Schrödinger operator on unbounded surface. Next, we consider a Schrödinger operator $\mathcal{L}$ on an unbounded surface M . The surface $M = M_{\text{int}} \cup M_{\text{ext}}$ is flat outside a compact region M_{int} and is shown along with the potential $V(x, y, z) = \chi_{M_{\text{int}}}(x, y, z) \exp(-5(x^2 + y^2))$ in Figure 10 (left), where $\chi_{M_{\text{int}}}(x, y, z)$ is the indicator function of the finite region. The infinite-surface Schrödinger problem on M may be reformulated into two coupled problems: a bounded-surface, variable-coefficient problem on M_{int} and an unbounded, flat, constant-coefficient problem on $M_{\text{ext}} = M \setminus M_{\text{int}}$. The latter problem can be solved using potential theory. We construct the surface operator on M_{int} using a variant of the method described in section 2 that uses impedance-to-impedance operators to avoid spurious resonances [37]. The boundary integral operators are constructed using the `chunkIE` package [2].

For $\alpha > 0$, we use the test functions

$$f(x, y, z) = \frac{c_\alpha \chi_{M_{\text{int}}}(x, y, z)}{[(x + 0.2)^2 + (y + 0.1)^2 + 1]^\alpha},$$

where c_α is a normalization constant so that $\|f\| = 1$. Figure 10 (right) shows the spectral measures computed using $\epsilon = 0.01$ and the 6th order kernel. The spectrum in this case is the whole positive half-line $[0, +\infty)$. We see that a larger value of α leads to slower decay of the spectral measure, just as in the case of the free Laplacian on $\mathbb{R}^2$. We also observe the excellent localization of the smooth measure to $\mathbb{R}_{\geq 0}$.

5. Conclusions. The `SurfSpec` methods developed in this paper provide robust, accurate algorithms for computing spectral properties of differential operators on surfaces; the methods are based on the resolvent and form part of a “solve-then-discretize” framework. The algorithms are flexible, come with error control and convergence guarantees and can be applied to a wide variety of surfaces, PDEs, and spectral problems. The power of these algorithms has been demonstrated via examples of Laplace–Beltrami and Schrödinger operators on complicated surfaces, such as the triceratops or an unbounded manifold, and internal waves on the torus.

`SurfSpec` can find application in diverse fields of mathematics. Beyond applications such as spectral shape analysis or data representation, it can be used to inform and prove conjectures in spectral geometry. For example, by combining with numerical optimization algorithms, we might test and formulate conjectures in shape optimization and spectral theory [28, 46]. The framework for computing spectral measures

using high-order kernels can be extended to computing the vector-valued functional calculus of operators on manifolds [73, Sec. VII.1]. This in turn can be used to solve linear evolutionary PDEs [27]; more complicated evolutionary PDEs can be solved by combining this with splitting methods [60]. The contour integral methods developed in this paper could also be utilized for system realization and control for dynamics on surfaces [11].

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